23 FEBRERO 2015

1) $H_c(s) = \dfrac{s+\alpha}{(s+\alpha)^2+\beta}$; $\alpha, \beta \in \mathbb{R}$

CEROS:

$s = -\alpha$

POLOS:

$s^2 + 2\alpha s + \alpha^2 + \beta = 0$

$s^2(1) + s(2\alpha) + (\alpha^2+\beta) = 0$

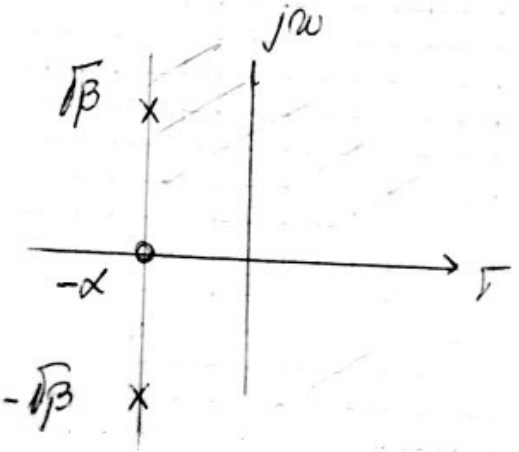
$$\frac{-2\alpha \pm \sqrt{4\alpha^2 - 4(\alpha^2+\beta)}}{2} = -\frac{2\alpha \pm 2\sqrt{-\beta}}{2} = -\alpha \pm \sqrt{-\beta}$$

$$= -\alpha \pm j\sqrt{\beta} = p_{1,2}$$

$\alpha > 0 \wedge \beta > 0$

CERO: $s = -|\alpha|$

POLOS: $p_{1,2} = -\alpha \pm j\sqrt{\beta}$



ROC: $\mathrm{Re}\{s\} > -\alpha$

$\sigma > -\alpha$

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$\alpha > 0,\ \beta < 0$

Cero: $s = -\alpha$

Polos $p_{1,2} = \alpha \pm \sqrt{|\beta|}$

↳ YA NO SON COMPLEJOS CONJUGADOS

•) $\alpha > \beta$

$p_1 = \alpha + \sqrt{|\beta|} > 0$ NO PUEDE SER ESTABLE!

$p_2 = \alpha - \sqrt{|\beta|} > 0$

•) $\beta > \alpha$

$p_1 = \alpha + \sqrt{|\beta|} > 0$ NO PUEDE SER ESTABLE

$\alpha < 0,\ \beta > 0$

Polos $p_{1,2} = |\alpha| \pm j\sqrt{\beta}$

↳ NO PUEDE SER ESTABLE

$\alpha < 0,\ \beta < 0$

Polos: $p_{1,2} = |\alpha| \pm \sqrt{|\beta|}$

↳ NO PUEDE SER ESTABLE

$\Rightarrow$ $\boxed{\alpha > 0 \wedge \beta > 0}$

$\boxed{\text{ROC: } \mathrm{Re}\{s\} > -\alpha}$

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b) $$H_c(s) = \frac{s+\alpha}{(s+\alpha-j\sqrt{\beta})(s+\alpha+j\sqrt{\beta})} = \frac{A}{(s+\alpha-j\sqrt{\beta})} + \frac{B}{(s+\alpha+j\sqrt{\beta})}$$

$$A = (s+\alpha-j\sqrt{\beta})\, H_c(s)\Big|_{s=-\alpha+j\sqrt{\beta}}$$

$$A = \frac{s+\alpha}{(s+\alpha+j\sqrt{\beta})}\Big|_{s=-\alpha+j\sqrt{\beta}} = \frac{-\alpha+j\sqrt{\beta}+\alpha}{-\alpha+j\sqrt{\beta}+\alpha+j\sqrt{\beta}}$$

$$A = \frac{j\sqrt{\beta}}{2j\sqrt{\beta}} = 1/2.$$

$$B = (s+\alpha+j\sqrt{\beta})\, H_c(s)\Big|_{s=-\alpha-j\sqrt{\beta}} = \frac{s+\alpha}{(s+\alpha-j\sqrt{\beta})}\Big|_{s=-\alpha-j\sqrt{\beta}}$$

$$B = \frac{-\alpha-j\sqrt{\beta}+\alpha}{-\alpha-j\sqrt{\beta}+\alpha-j\sqrt{\beta}} = 1/2$$

$$H_c(s) = \frac{1/2}{s+\alpha-j\sqrt{\beta}} + \frac{1/2}{s+\alpha+j\sqrt{\beta}} \quad ; \quad Re\{s\} > -\alpha$$

$$e^{-kt}u(t) \rightarrow \frac{1}{s+k}$$

$$h_s(t) = \left\{\frac{1}{2}e^{-(\alpha+j\sqrt{\beta})t} + \frac{1}{2}e^{-(\alpha-j\sqrt{\beta})t}\right\}u(t)$$

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$h_d[m] = h_c(m\bar{T})$, $\bar{T} > 0$

$$h_d[m] = \left\{ \frac{1}{2} e^{-(\alpha + j\sqrt{\beta})m\bar{T}} + \frac{1}{2} e^{-(\alpha - j\sqrt{\beta})m\bar{T}} \right\} u[m\bar{T}]$$

$$h_d[m] = \left\{ \frac{1}{2} e^{-(\alpha + j\sqrt{\beta})m\bar{T}} + \frac{1}{2} e^{-(\alpha - j\sqrt{\beta})m\bar{T}} \right\} u[m]$$

$$\alpha^m u[m] \longrightarrow \frac{1}{1 - \alpha z^{-1}} \quad ; \quad |z| > |\alpha|$$

$$H_d(z) = \frac{1/2}{1 - e^{-(\alpha + j\sqrt{\beta})\bar{T}} z^{-1}} + \frac{1/2}{1 - e^{-(\alpha - j\sqrt{\beta})\bar{T}} z^{-1}}$$

Polos:

$$p_1 = e^{-(\alpha + j\sqrt{\beta})\bar{T}} \quad ; \quad |p_1| = e^{-\alpha\bar{T}}$$

$$p_2 = e^{-(\alpha - j\sqrt{\beta})\bar{T}} \quad ; \quad |p_2| = e^{-\alpha\bar{T}}$$

$\Rightarrow$ ROC: $|z| > e^{-\alpha\bar{T}}$

$\left.\begin{array}{l} \alpha > 0 \\ \bar{T} > 0 \end{array}\right\}$ SISTEMA CAUSAL Y ESTABLE

•) PARA CUALQUIER VALOR DE $\bar{T}$ ES CAUSAL Y ESTABLE

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3) $H(e^{jw}) = \left(\frac{1}{2} + \cos(4w)\right) e^{-j4w}$

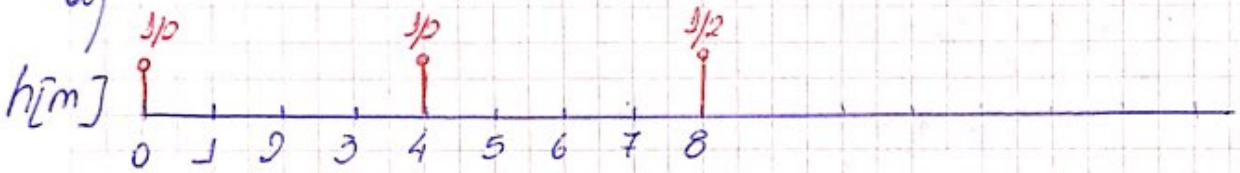
$$H(e^{jw}) = \left(\frac{1}{2} + \frac{e^{4jw} + e^{-4jw}}{2}\right) e^{-j4w}$$

$$H(e^{jw}) = \frac{1}{2} e^{-j4w} + \frac{1}{2} + \frac{1}{2} e^{-j8w}$$

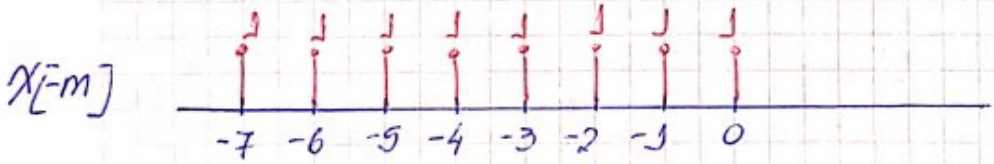
$$\Rightarrow h[m] = \frac{1}{2}\delta[m] + \frac{1}{2}\delta[m-4] + \frac{1}{2}\delta[m-8] \rightarrow Long = 9$$

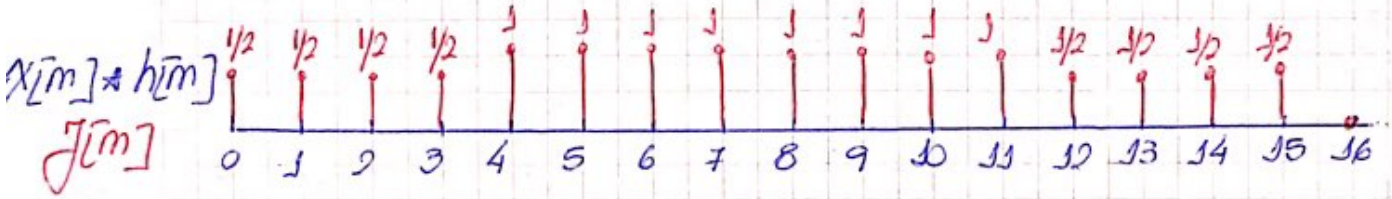
$$x[m] = 1 \; ; \; m = 0, \cdots, 7 \rightarrow Long = 8$$

a)









$h[m] \rightarrow L = 9$

$x[m] \rightarrow ? = 8$

$x[m] * h[m] = L + ? - 1$

$= 16.$

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b) EL VALOR DE N DEBE SER TAL QUE $N \geq L+P-1$.

$N \geq 16$

OPTANDO POR N=16 SUCEDE QUE LA CONVOLUCIÓN LINEAL Y CIRCULAR COINCIDEN.

$$\rightarrow H(e^{j\omega}) = \frac{1}{2} + \frac{1}{2}e^{-j4\omega} + \frac{1}{2}e^{-j8\omega}$$

LA DFT SURGE DE MUESTREAR LA TRANSFORMADA DE FOURIER:

$$H[K] = H(e^{j\omega})\Big|_{\omega=\frac{2\pi K}{16}}$$

$$H[K] = \frac{1}{2} + \frac{1}{2}e^{-j4\cdot\frac{2\pi K}{16}} + \frac{1}{2}e^{-j8\cdot\frac{2\pi K}{16}}$$

$$H[K] = \frac{1}{2} + \frac{1}{2}e^{-j\frac{\pi}{2}K} + \frac{1}{2}e^{-j\pi K} \quad ; \quad K \in [0, 15].$$

$$x[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \cdots + \delta[n-7]$$

$$X(e^{j\omega}) = 1 + e^{-j\omega} + e^{-j2\omega} + e^{-j3\omega} + e^{-j4\omega} + e^{-j5\omega} + e^{-j6\omega} + e^{-j7\omega}$$

$$X[K] = 1 + e^{-j\frac{2\pi K}{16}} + e^{-j\frac{2\cdot 2\pi K}{16}} + e^{-j\frac{3\cdot 2\pi K}{16}} + e^{-j\frac{4\cdot 2\pi K}{16}} + e^{-j\frac{5\cdot 2\pi K}{16}} + e^{-j\frac{6\cdot 2\pi K}{16}} + e^{-j\frac{7\cdot 2\pi K}{16}}$$

$$X[K] = 1 + e^{-j\frac{\pi}{8}K} + e^{-j\frac{\pi}{4}K} + e^{-j\frac{3\pi}{8}K} + e^{-j\frac{\pi}{2}K} + e^{-j\frac{5\pi}{8}K} + e^{-j\frac{3\pi}{4}K} + e^{-j\frac{7\pi}{8}K}$$

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